Activity 2.3: Login and Register page routings

- Login page

`/login`: Authenticates user credentials and initiates a session upon successful login.

```python
@app.get("/login", response_class=HTMLResponse)
async def login_page(request: Request):
    """Serve the login page"""
    # Check if user is already logged in
    try:
        token = await get_token(request)
        if token:
            user = await get_current_user(request, token)
            if user:
                return RedirectResponse(url="/dashboard", status_code=status.HTTP_302_FOUND)
    except Exception:
        pass

    return templates.TemplateResponse("login.html", {"request": request})
```

- Register Page

`/register`: Handles new user registration by accepting and securely storing user credentials.

```python
@app.get("/register", response_class=HTMLResponse)
async def register_page(request: Request):
    """Serve the registration page"""
    # Check if user is already logged in
    try:
        token = await get_token(request)
        if token:
            user = await get_current_user(request, token)
            if user:
                return RedirectResponse(url="/dashboard", status_code=status.HTTP_302_FOUND)
    except Exception:
        pass

    return templates.TemplateResponse("register.html", {"request": request})
```

- Logout Page

`/logout`: Terminates the current user session and redirects to the login page.

```python
@app.post("/logout")
async def logout(request: Request):
    """Logout user by blacklisting their token and clearing session"""
    token = await get_token(request)

    if token:
        # Add token to blacklist
        blacklisted_tokens.add(token)

        try:
            # Extract username from token to remove from active sessions
            payload = jwt.decode(token, SECRET_KEY, algorithms=[ALGORITHM])
            username = payload.get("sub")
            if username and username in active_sessions:
                del active_sessions[username]
        except JWTError:
            pass

    # Use RedirectResponse instead of JSONResponse for proper redirection
    response = RedirectResponse(url="/login", status_code=status.HTTP_302_FOUND)
    response.delete_cookie(key="access_token")
```

Activity 2.4: Ensure modular code structure

- `/token`: Issues a secure JWT token after successful authentication, enabling authorized access to protected API routes.

```python
@app.post("/token", response_model=Token)
async def login_for_access_token(form_data: OAuth2PasswordRequestForm = Depends()):
    """Login endpoint to get access token"""
    user = authenticate_user(users_db, form_data.username, form_data.password)
    if not user:
        raise HTTPException(
            status_code=401,
            detail="Incorrect username or password",
            headers={"WWW-Authenticate": "Bearer"},
        )

    # Create access token
    access_token_expires = timedelta(minutes=ACCESS_TOKEN_EXPIRE_MINUTES)
    access_token = create_access_token(
        data={"sub": user.username}, expires_delta=access_token_expires
    )

    # Create or update session for the user
    existing_user_data = {}
    if user.username in active_sessions:
        existing_user_data = active_sessions[user.username].user_data
        # If there's an old token, blacklist it
        old_token = active_sessions[user.username].token
        blacklisted_tokens.add(old_token)

    active_sessions[user.username] = UserSession(
        username=user.username,
        login_time=datetime.utcnow(),
        last_activity=datetime.utcnow(),
        token=access_token,
        user_data=existing_user_data
    )

    # Return response with cookie
    response = JSONResponse(content={"access_token": access_token, "token_type": "bearer"})

    response.set_cookie(
        key="access_token",
        value=access_token,  # Store token directly without Bearer prefix
        httponly=True,
        max_age=ACCESS_TOKEN_EXPIRE_MINUTES * 60,
        samesite="lax"
    )

    return response
```

- `/session-info`: Retrieves metadata about the current user session, such as user ID and login status.
- `/session-data`: Returns detailed session-specific data used for personalization and recommendation tracking.

```python
@app.get("/session-info")
async def get_session_info(request: Request, current_user: UserInDB = Depends(get_current_active_user)):
    """Get current user's session information"""
    if current_user.username in active_sessions:
        session = active_sessions[current_user.username]
        return {
            "username": session.username,
            "login_time": session.login_time,
            "last_activity": session.last_activity,
            "session_duration": (datetime.utcnow() - session.login_time).total_seconds() // 60,  # in minutes
            "user_data": session.user_data
        }
    else:
        raise HTTPException(status_code=404, detail="No active session found")

@app.post("/session-data")
async def update_session_data(
    data: Dict[str, Any],
    request: Request,
    current_user: UserInDB = Depends(get_current_active_user)
):
    """Update user's session data"""
    if current_user.username in active_sessions:
        active_sessions[current_user.username].user_data.update(data)
        active_sessions[current_user.username].last_activity = datetime.utcnow()
        return {"message": "Session data updated", "data": active_sessions[current_user.username].user_data}
    else:
        raise HTTPException(status_code=404, detail="No active session found")
# Fix for the /home-budget endpoint to properly handle the form submission
```

MILESTONE 3: Backend – FastAPI Integration (main.py)

This milestone focuses on implementing the FastAPI-powered backend for **PocketSmart AI**. It acts as the bridge between the frontend interface and the Gemini-based AI recommendation engine defined in gemini_utils.py. The backend is designed with modular architecture, providing clean and maintainable API endpoints for planners across Home, Party, and Jewelry domains.It also includes features like CORS setup for cross-origin requests, session handling for personalized user experience, and structured input/output models to ensure smooth AI interaction and reliable frontend rendering.

- **Activity 3.1: Define FastAPI Routes**
  - Home Planner (`/generate-home`): Accepts home-related preferences and budget to return tailored product suggestions.

```python
@app.post("/home-budget")
async def plan_home_budget(
    budget_input: HomeBudgetInput,
    request: Request,
    current_user: UserInDB = Depends(get_current_active_user)
):
    """Generate home budget recommendations"""
    # Store last budget planning in session data
    if current_user.username in active_sessions:
        active_sessions[current_user.username].user_data["last_home_budget"] = {
            "timestamp": datetime.utcnow().isoformat(),
            "budget": budget_input.total_budget,
            "requirements": {
                "lights": budget_input.num_lights,
                "fans": budget_input.num_fans,
                "furniture": budget_input.num_furniture,
                "dining_tables": budget_input.num_dining_tables
            }
        }

    # Get recommendations
    result = get_home_recommendations(budget_input)

    # Save to history
    save_to_history(
        username=current_user.username,
        recommendation_type="home",
        input_data=budget_input.dict(),
        result=result
    )

    return result
```

- Party Planner (`/generate-party`): Processes party details and guest count to recommend venue, food, and decoration items.